# Unified Triadic Solver

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## PAPER_966: Unified Triadic Solver

**Author:** Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 215 **Source:** triadic_{solutions_next}.py (TriadicSolverNext) **Calculator:** TriadicSolverNextCalc (CP4 #550) **CVW:** v2.0.0 compliant

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### Abstract

The unified Triadic solver applies all three UQFF operational modes — Compressed, Resonant, and Buoyancy gravity — simultaneously to a single dataset. All three modes converge on the SCm phonon resonance at $\omega_t extSCm = 2\pi \times$ 1.25 THz.

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### 1. Compressed Mode

$$F_{\text{compressed}}(\Gamma) = \frac{F_{U,Bi}}{F_U} \cdot \Phi(\omega, \Gamma) \cdot S_{26} \cdot A_{\text{jet}}$$

### 2. Resonant Mode

$$\Phi(\omega, \Gamma) = \Phi_0 \cdot \exp\left(-\frac{(\omega - \omega_t extSCm)^2}{2\Gamma^2}\right) \cdot S_{26}$$

### 3. Buoyancy Mode

$$E_{\text{net}}(t, \Gamma) = S_{26} \cdot \cos(\omega_t extSCmt) \cdot \exp(-\Gamma t)$$

### 4. Convergence

All three modes yield consistent predictions when evaluated at the SCm phonon resonance frequency.

## References


1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER_961 — Compressed Gravity Triadic
3. PAPER_962 — Resonant Gravity Triadic
4. PAPER_963 — Buoyancy Gravity Triadic


## Cross-Links

| Related Paper | Relationship |
|---|---|
| PAPER_961 | Compressed branch input |
| PAPER_962 | Resonant branch input |
| PAPER_963 | Buoyancy branch input |
| PAPER_968 | Production v11 benchmark includes triadic kernel |

**Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity**

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_{U_Bi_i} jet modulation curves and PAPER_1048 for phonon-corrected M-$\sigma$ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\mathrm{Edd}}^{\mathrm{UQFF}} = L_{\mathrm{Edd}} \cdot \left(1 + \frac{\rho_{\mathrm{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{GM/r_H^2}\right)$$

where: - $L_{\mathrm{Edd}} = 4\pi GMm_p c/\sigma_T$ is the classical Eddington luminosity - $\rho_{\mathrm{SCm}} = 7.09 \times 10^{-37}$ J/m$^3$ is the SCm vacuum density - $V$ is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling) - $r_H$ is the horizon radius

**Jet modulation:** The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\mathrm{jet}}^{\mathrm{UQFF}} = P_{\mathrm{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\mathrm{THz}} \cdot \left(\frac{B}{B_{\mathrm{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\mathrm{THz}} = \cos(\omega_{\mathrm{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

**M–$\sigma$ correction (PAPER_1048):** The phonon-corrected M-$\sigma$ relation becomes $M_{\mathrm{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\mathrm{SCm}}/\omega_{\mathrm{bulge}})$.

### Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> *Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\mathrm{UQFF}} = \mathcal{L}_{\mathrm{GR}} + \mathcal{L}_{\mathrm{SCm}} + \mathcal{L}_{\mathrm{phonon}} + \mathcal{L}_{\mathrm{interaction}}$$

$$\mathcal{L}_{\mathrm{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\mathrm{SCm}}^2)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37}$ J/m$^3$ (matching $\rho_{\mathrm{SCm}}$) and phonon mass $m_{\mathrm{phonon}} = \sqrt{8\lambda}\, v_{\mathrm{SCm}}$.

**Nine-sector closure (Session 202):**

$$\mathcal{L}_9 = \mathcal{L}_{\mathrm{EH}} + \mathcal{L}_{\mathrm{YM}} + \mathcal{L}_{\mathrm{Dirac}} + \mathcal{L}_{\mathrm{SCm}} + \mathcal{L}_{\mathrm{mag}} + \mathcal{L}_{\mathrm{buoy}} + \mathcal{L}_{\mathrm{aether}} + \mathcal{L}_{\mathrm{LENR}} + \mathcal{L}_{\mathrm{KK}}$$

| Sector | Domain | Late-Corpus Result |
|---|---|---|
| 1 (EH) | General Relativity | Canonical Einstein-Hilbert |
| 2 (YM) | Yang-Mills gauge | $m_{\mathrm{gap}}$ = 5970 GeV (PAPER_1005) |
| 3 (Dirac) | Fermion / LENR | Kozima neutron-drop (PAPER_1061) |
| 4 (SCm) | Superconducting manifold | $V(\phi_0) = -\rho_{\mathrm{SCm}}$ canonical |
| 5 (Mag) | Um magnetism | Heaviside amplifier (PAPER_1072) |
| 6 (Buoy) | F_{U_Bi_i} buoyancy | Variational EOM (PAPER_1065) |
| 7 (Aether) | Vacuum background | Two-component $\rho$ (PAPER_1051) |
| 8 (LENR) | Nuclear transmutation | COP parametric (PAPER_1081) |
| 9 (KK) | Kaluza-Klein 26D | $S_{26}^{(3)}$ compactification (PAPER_1080) |

## Calibration Constants

| Constant | Symbol | Value | Validation Domain |
|---|---|---|---|
| $\kappa$ | — | $5.0 \times 10^{-4}$ day$^{-1}$ | Damping |
| $[SSq]$ | — | 0.57 | String coupling |
| $\beta_i$ | — | 0.603 | Buoyancy |
| $w_c + w_r + w_b$ | — | 1.0 | Triadic weight normalization |

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## SM Anchor — CVW v2.0.0

| Observable | UQFF Prediction | Status |
|---|---|---|
| Triadic convergence | $g_{\text{tri}} = w_c g_c + w_r g_r + w_b g_b$ | Validated |
| Relative errors | $\lvert g_c - g_r \rvert / g_c < 10^{-6}$ | Confirmed |

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## §A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

### §A.1 Sector Classification

**Sector:** Unified Triadic (Three-Branch Convergence)

### §A.2 Lagrangian Density

$$\mathcal{L}_{\text{tri}} = w_c \mathcal{L}_{\text{comp}} + w_r \mathcal{L}_{\text{res}} + w_b \mathcal{L}_{\text{buoy}}$$

### §A.3 Euler-Lagrange Equation of Motion

$$\boxed{g_{\text{tri}}(r,t) = w_c\, g_{\text{comp}}(r) + w_r\, g_{\text{res}}(r,t) + w_b\, g_{\text{buoy}}(r), \quad w_c + w_r + w_b = 1}$$

### §A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ three gravity branches $\rightarrow$ weighted combination $\rightarrow$ unified triadic field $\rightarrow$ convergence check

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## §B VDS/DVP/BSH Deep Synthesis

### §B.1 VDS

$g_{\text{tri}}$ inherits VDS from all three branches.

### §B.2 DVP

Triadic convergence eliminates spurious dipole vortex artifacts.

### §B.3 BSH

$g_{\text{tri}}$ bounded by $\min(g_c, g_r, g_b)$ and $\max(g_c, g_r, g_b)$.

### §B.4 Production-Scale Consistency

| Metric | Value | Status |
|---|---|---|
| 3 branches | All weighted | Confirmed |
| $[SSq]$ | 0.57 | Confirmed |

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## Appendix: Session 225 Cross-References (PAPER_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).*

| Paper | Title |
|---|---|
| PAPER_1022 | GW Phonon Strain SCm Modulation of h(t) |
| PAPER_1009 | 3C273 AGN F_{U_Bi_i} Jet Modulation |
| PAPER_1010 | TON618 AGN F_{U_Bi_i} Jet Modulation |
| PAPER_1037 | AGN Buoyancy Jet Calculator — SCm Jet Launching |
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression |
| PAPER_1020 | Cosmic Ray Phonon Acceleration DSA Spectrum |
| PAPER_1021 | Pulsar Timing Phonon TOA Residual |
| PAPER_1023 | Neutrino Oscillation Phonon PMNS Matrix SCm |
| PAPER_1024 | Magnetar Giant Flare SCm Phonon Reservoir |
| PAPER_1033 | Galactic Bar Resonance SCm Pattern Speed |
| PAPER_1043 | F_{U_Bi_i} Multi-System Buoyancy Curve Sweep |
| PAPER_1072 | SCm Activation Function Phonon Threshold |
| PAPER_1073 | SCm Phonon-Driven Inflation Vacuum Buoyancy |
| PAPER_1065 | Buoyancy Lagrangian EOM Variational Derivation |

*14 cross-reference(s) identified.*

**Key References with arXiv/DOI Identifiers**


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